Mutual Fund Performance Analysis Dashboard

Data Analyst Project Assignment

Project Title

Mutual Fund Performance Analysis Dashboard using Python, Excel, and Power BI

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1. Project Overview

The objective of this project is to analyze and rank mutual funds based on their performance, expense ratio, and risk-adjusted returns. Students will use Python for data analysis, Excel for validation and formatting, and Power BI for interactive dashboard development.

This project simulates a real-world financial analytics scenario where investors need data-driven insights to select suitable mutual funds.

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2. Business Problem

Investors often face difficulties selecting the right mutual fund due to the availability of hundreds of investment options.

Your task is to:

- Analyze 500+ mutual funds
- Compare returns, risk, and expenses
- Develop a ranking system
- Identify top-performing funds
- Build an interactive dashboard for decision-making

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3. Learning Objectives

After completing this project, students will be able to:

- Perform Exploratory Data Analysis (EDA)
- Handle missing values and data cleaning
- Apply feature engineering techniques
- Normalize data using MinMaxScaler
- Create weighted scoring systems
- Rank mutual funds based on performance
- Build professional Power BI dashboards
- Create DAX measures and KPIs

• Generate actionable business insights

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4. Tools & Technologies

Tool	Purpose
Python	Data Analysis & Ranking
Pandas	Data Manipulation
NumPy	Numerical Operations
Scikit-Learn	Data Scaling
Excel	Validation & Formatting
Power BI	Dashboard Development
Power Query	Data Transformation
DAX	Measures & KPIs

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5. Dataset Information

Dataset Name: Mutual Fund.csv

Expected Columns:

• Fund Name
• Category
• AMC Name
• Expense Ratio
• Returns 1Y
• Returns 3Y
• Returns 5Y
• Sharpe Ratio
• Sortino Ratio
• Alpha
• Beta
• Fund Size
• SIP Amount
• Lumpsum Amount

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6. Python Tasks

Task 1: Data Loading

- Import dataset using Pandas
- Display first 10 records
- Check data types

Deliverables

- Dataset overview
- Column information

Task 2: Exploratory Data Analysis

Perform:

- Shape of dataset
- Summary statistics
- Unique categories
- Null value analysis

Expected Functions

```
df.head()
df.info()
df.describe()
df.isnull().sum()
df.nunique()
```

Task 3: Data Cleaning

Perform:

- Replace invalid values
- Handle missing values
- Convert columns into numeric format

Deliverables

Clean dataset ready for analysis

Task 4: Feature Engineering

Select important metrics:

- Expense Ratio
- Returns 1Y
- Returns 3Y
- Returns 5Y
- Sharpe Ratio
- Sortino Ratio
- Alpha
- Beta

Task 5: Data Normalization

Use MinMaxScaler.

```
from sklearn.preprocessing import MinMaxScaler
```

Scale values between 0 and 1.

Invert:

- Expense Ratio
- Beta

Reason: Lower values are preferred.

Task 6: Composite Score Calculation

Create a weighted scoring model.

Suggested Weights:

Metric	Weight
Returns 1Y	20%
Returns 3Y	20%
Returns 5Y	20%
Expense Ratio	15%
Sharpe Ratio	10%
Sortino Ratio	5%

Metric	Weight
Alpha	5%
Beta	5%

Formula:

Composite Score = Σ (Normalized Metric × Weight)

Task 7: Fund Ranking

Rank funds based on Composite Score.

```
df['Rank'] = df['Composite Score'].rank(ascending=False)
```

Identify:

- Top 10 Funds
- Top 20 Funds
- Top 30 Funds

Export results to Excel.

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7. Excel Tasks

Perform the following:

Data Validation

- Verify missing values
- Check duplicates
- Format percentage columns

Conditional Formatting

Highlight:

- Top 10 funds
- Highest returns
- Lowest expense ratios

Dashboard Support Tables

Create:

- Category Summary

- AMC Summary
- Return Analysis

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8. Power BI Tasks

Data Import

Load cleaned dataset.

Power Query Tasks

Perform:

- Column Renaming
- Data Type Conversion
- Null Handling
- Percentage Formatting

DAX Measures

Create:

Average 1-Year Return

```
Avg Return =
AVERAGE(MutualFund[returns_1yr])
```

Average Expense Ratio

```
Avg Expense Ratio =
AVERAGE(MutualFund[expense_ratio])
```

Composite Score

Create DAX equivalent.

Fund Rank

Create ranking measure.

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9. Dashboard Requirements

Design a professional dashboard containing:

KPI Cards

- Average Return
- Average Expense Ratio
- Average Sharpe Ratio
- Total Fund Size

Visualizations

Bar Chart

Top 30 Mutual Funds

Column Chart

Category-wise Returns

Scatter Plot

Risk vs Return

X Axis: Beta

Y Axis: Return

Pie Chart

Fund Category Distribution

Table Visual

Top Funds with:

- Fund Name
- Category
- Return
- Expense Ratio
- Composite Score
- Rank

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10. Interactivity Requirements

Add slicers for:

- Fund Category
- AMC
- Return Period
- Risk Category

Enable:

- Cross Filtering
- Drill Through
- Tooltips

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11. Business Insights Questions

Students must answer:

1. Which category provides the highest returns?
2. Which fund has the lowest expense ratio?
3. Which AMC manages the highest assets?
4. Which funds provide the best risk-adjusted returns?
5. Is there a relationship between expense ratio and returns?
6. Which category is most suitable for long-term investment?
7. Which Top 10 funds should investors consider?

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12. Deliverables

Students must submit:

Python Files

- .ipynb Notebook
- Source Code

Excel Files

- Cleaned Dataset
- Validation Workbook

Power BI Files

- .pbix File

Project Report

- Business Problem
- Methodology
- Findings
- Recommendations

Dashboard Screenshots

- Full Dashboard
- Insights Page

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13. Evaluation Rubric

Criteria	Marks
Data Cleaning	15
EDA	15
Feature Engineering	15
Scoring Model	15
Power BI Dashboard	20
Insights & Recommendations	10
Documentation	10
Total	100

Expected Outcome

Students will build a complete end-to-end financial analytics solution capable of identifying top-performing mutual funds using Python, Excel, and Power BI while generating actionable investment insights through interactive dashboards.